ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **SUMMER SEMESTER, 2023-2024**

DURATION: 1 HOUR 30 MINUTES **FULL MARKS: 50**

CSE 4615: Wireless Networks

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Explain the concept of ***channel capacity*** in wireless communication and identify the key factors that influence it. 4 (CO1) (PO2)

Solution:

Channel Capacity : The maximum rate at which data can be transmitted over a given communication path, or channel, under given conditions. **Formula (Shannon Capacity)**

$$C = B \log_2(1 + \text{SNR})$$

Key Factors Influencing Channel Capacity

- **Bandwidth (B):** More bandwidth → higher data rate (but spectrum is limited).
- **SNR (Signal-to-Noise Ratio):** Stronger signal vs. noise → better capacity.
- **Interference:** More interference → worse capacity.
- **Fading & Multipath:** Signal fluctuations → reduce performance.
- **Modulation Scheme:** Higher-level modulation = more bits per symbol, but requires high SNR.
- **Coding Efficiency:** Good error correction helps approach the Shannon limit.
- **MIMO (Multiple Antennas):** Send data in parallel streams → big capacity boost.

b) Clarify the broadcast limitations at wireless links by answering the following questions:

i. Why is an **explicit acknowledgment** from the receiving station to the transmitting station necessary in wireless communication, but typically not required in wired networks? 2.5 (CO2) (PO2)

Solution:

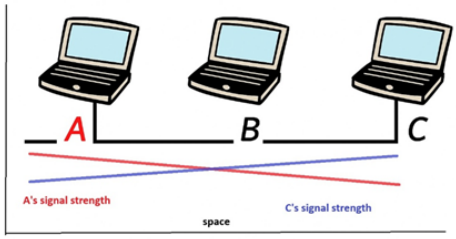


Figure 1: Received Signal Strength in Wired Networks

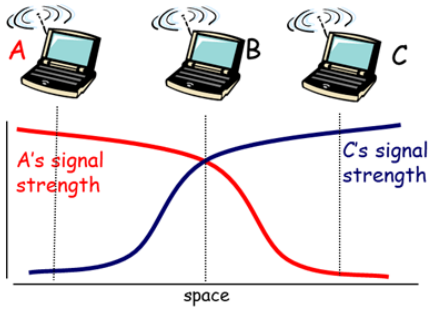


Figure 2: Received Signal Strength in Wireless Networks

Wireless Links

- The channel is more error-prone (noise, fading, interference).
- Signal attenuation is very high.
- The channel condition at the senders location is not the same as at the receivers location.
- Therefore, the sender cannot determine, by staying at its own location, whether the packet has been successfully received.

Wired Links

- The channel is stable with very low error rates.
- Signal attenuation is very low.
- The channel condition at the senders location is same as at the receivers location.
- Therefore, the sender can determine, by staying at its own location, whether the packet has been successfully received.
- Collisions are detectable via CSMA/CD using full-duplex radios / dual antennas (e.g., Ethernet can sense & resend).
- Thus, explicit ACK is not needed at the link layer.

Core Reason In wireless, the sender cannot rely on its own sensing to infer the receiver's state → explicit ACK is essential.

ii. Why does the ***hidden terminal problem*** occur in multi-hop wireless networks, but not in wired networks? 2.6 (CO2) (PO2)

Solution:

Wireless Links (Multi-hop):

- As the signal attenuation is very high, the channel condition at the senders location is not the same as at the receivers location.
- A sender may sense the channel as idle, while a hidden node (out of its range) may still interfere at the receiver.
- Therefore, the sender cannot determine, by staying at its own location, whether the receivers channel is actually free.
- This mismatch in channel conditions leads to the hidden terminal problem.

Wired Links:

- As the signal attenuation is very low, the channel condition at the senders location is essentially the same as at the receivers location.
- Every node attached to the wire observes almost the same medium state.
- Collisions are detectable (e.g., CSMA/CD in Ethernet).
- Thus, no hidden terminal problem arises.

c) ***6G* networks** are expected to deliver data rates 10 to 100 times faster than *5G*, with theoretical peak speeds reaching up to **1 *Tbps***. Briefly discuss one key technological concept anticipated to support such ultra-high-speed wireless communication. 3

Solution:

AI-Integrated Semantic Communication in 6G

- **Key Concept:** One key technological concept for 6G ultra-high-speed communication is *AI-integrated Semantic Communication*.
- **Traditional vs. Semantic:** Instead of transmitting raw bits, semantic communication focuses on transmitting the *intended meaning* of the message.
- **Role of AI:** By integrating AI, the system can understand, compress, and reconstruct information intelligently.
- **Benefits:**
  - Drastically reduces redundant data.
  - Improves efficiency.
  - Supports Tbps-level data rates required in 6G.

d) In a wireless communication system, a transmitter emits a signal with a transmit power (P_{tx}) of 100 mW. Due to attenuation in the wireless channel, the received signal power (P_{rx}) at a distance of 900 meters is measured to be 0.1 μW. The receiver has a sensitivity threshold ($P_{rx\ sensitivity}$) of 80 dBm, meaning it can only correctly decode signals stronger than this threshold.

To convert power from *milliwatts (mW)* to *decibel-milliwatts (dBm)*, use the formula in equation 1:

$$P_{dBm} = 10 \cdot \log_{10} P_{mW} \tag{1}$$

The *maximum tolerable path loss* can be calculated using formula in equation 2:

$$\textit{Maximum Tolerable Path Loss} = P_{\text{tx (dBm)}} - P_{\text{rx sensitivity (dBm)}} \tag{2}$$

i. Calculate the ***path loss*** in *decibels (dB)* experienced by the signal over the 900 meter wireless channel. 2.3

Solution:

Given:

$$P_{\text{tx}} = 100 \text{ mW}$$
$$P_{\text{rx}} = 0.1\ \mu\text{W} = 0.0001 \text{ mW}$$
$$\text{Receiver Sensitivity} = 80 \text{ dBm}$$

Step 1: Convert Transmit Power to dBm

$$P_{\text{tx (dBm)}} = 10 \cdot \log_{10}(100) = 10 \cdot 2 = \boxed{20 \text{ dBm}}$$

Step 2: Convert Received Power to dBm

$$P_{\text{rx (dBm)}} = 10 \cdot \log_{10}(0.0001) = 10 \cdot (-4) = \boxed{-40 \text{ dBm}}$$

Step 3: Compute Path Loss

$$\text{Path Loss (dB)} = P_{\text{tx (dBm)}} - P_{\text{rx (dBm)}} = 20 - (-40) = \boxed{60 \text{ dB}}$$

ii. Compute the ***maximum tolerable path loss*** for the system. Based on your result in 1.d)i, determine whether the receiver at 900 meters can function correctly. 2.3 (CO1) (PO2)

Solution:

Step 1: Compute Maximum Path Loss

$$\text{Max Path Loss} = P_{\text{tx (dBm)}} - P_{\text{rx sensitivity (dBm)}} = 20 - 80 = \boxed{-60 \text{ dB}}$$

Step 2: Compare with Actual Path Loss

$$\text{Actual Path Loss} = 60 \text{ dB} \quad \text{Max Allowable Path Loss} = -60 \text{ dB}$$
$$60 \text{ dB} > -60 \text{ dB}$$

Conclusion:

No, the receiver at 900 meters can not function correctly, since the actual path loss (60 dB) is higher than the maximum tolerable limit (-60 dB).

2. a) i. Discuss the communication requirements for each of the following applications in terms of approximate levels of ***maximum acceptable bit error probability, minimum latency,*** and ***required throughput*** (e.g., very low, low, high, very high): 2.3 (CO1) (PO2)

- ***Email***
- ***Voice calling over cellular network***
- ***Web browsing using mobile broadband***

Solution:

Application	Bit Error Probability	Latency	Throughput
Email	Very low ($< 10^{-9}$)	High (delay < 1 day acceptable)	Low (rate not critical)
Voice Calling	Low (< 0.1%)	Very low (delay < 100 ms)	Low (10–15 kbit/s)
Web Browsing	Low ($< 10^{-5}$)	Moderate (delay < few seconds)	Moderate (depends on content)

ii. Based on your analysis, critically evaluate whether a ***'one-size-fits-all'*** communication protocol and network design can efficiently support all of these applications. 1 (CO2) (PO2)

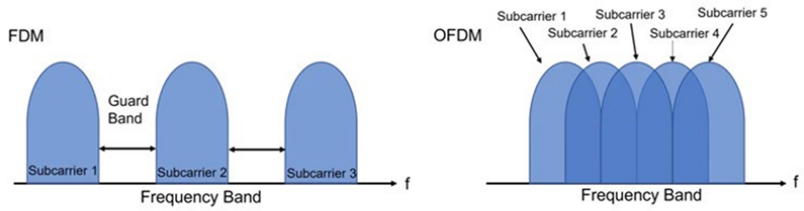
Solution:

Answer No, a one-size-fits-all protocol is inefficient. Instead:

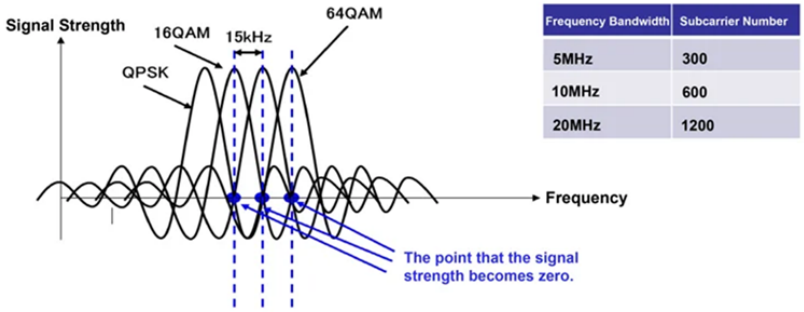
- Differentiated QoS mechanisms are required
- Application-specific network optimizations are needed
- Example: separate classes for real-time, delay-tolerant, and high-throughput services

b) Highlight the key similarities and differences between ***Orthogonal Frequency Division Multiplexing (OFDM)*** and ***Multiple-Input Multiple-Output (MIMO)*** in the context of modern wireless communication systems, and support your explanation with suitable real-world analogies. 4.6 (CO2) (PO2)

Solution:



(a) Conceptual diagram of multi-carrier transmission



Frequency Bandwidth	Subcarrier Number
5MHz	300
10MHz	600
20MHz	1200

(b) Subcarrier arrangement in OFDM

Figure 3: Orthogonal Frequency Division Multiplexing illustrations

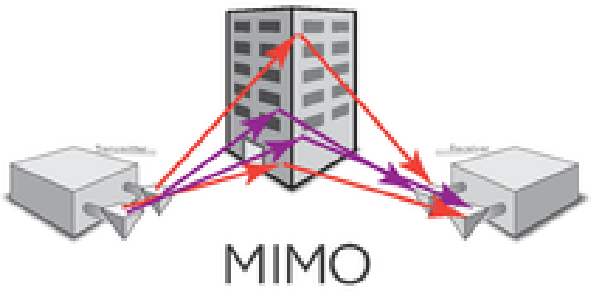


Figure 4: Multiple-Input Multiple-Output

Similarities

- Both aim to increase data rate and reliability.
- Both are used together in modern systems (e.g., Wi-Fi 6, 5G).
- Both support parallelism to handle high-speed data efficiently.

Differences

- **Transmission medium:**

- OFDM → Same antenna, many frequencies.
- MIMO → Many antennas, same frequency.

- **Focus:**
  - OFDM → Handles frequency interference (multipath).
  - MIMO → Handles spatial diversity / multipath.
- **Hardware need:**
  - OFDM → Can work with a single antenna.
  - MIMO → Needs multiple antennas.

Real-world Analogies

- **OFDM:** Like sending a message split across many radio stations (frequencies) at once.
- **MIMO:** Like sending different messages from different people (antennas) all at the same time following different physical path.

c) Consider a *Basic Service Set (BSS)* in an *IEEE 802.11 Wireless LAN (WLAN)*, consisting of three stations: *A, B,* and *C*. All stations use the ***Distributed Coordination Function (DCF)*** and are within mutual transmission range (i.e., each station can sense the others' transmissions). 8.8 (CO2) (PO2)

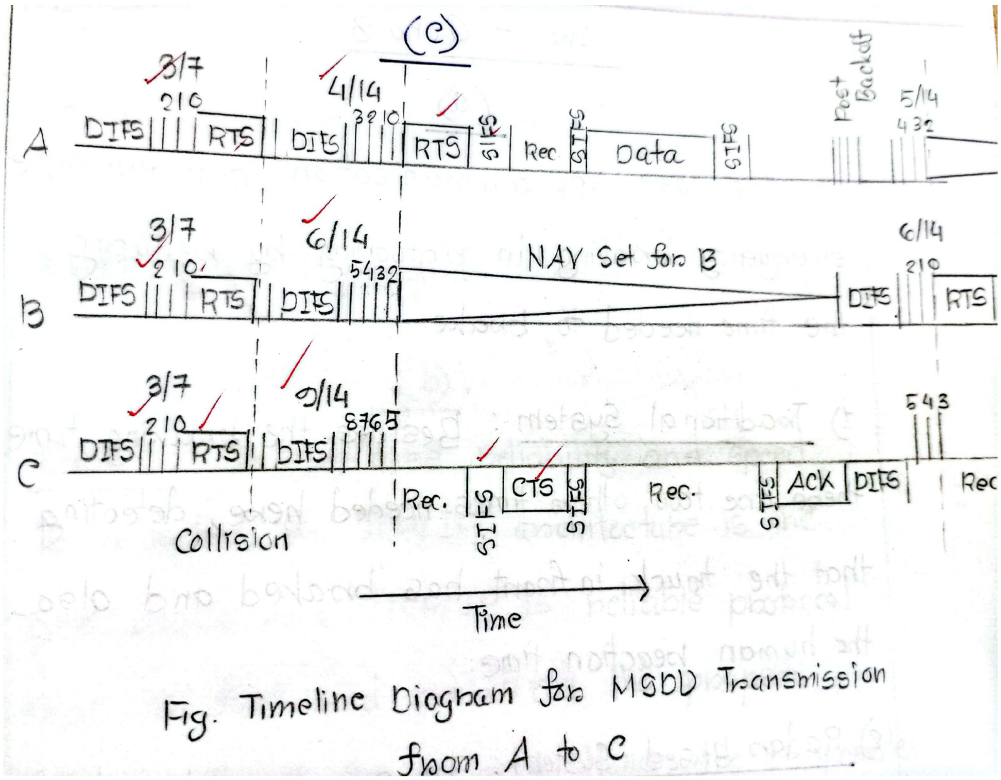
Suppose Station *A* attempts to transmit a *MAC Service Data Unit (MSDU)* to Station *C*. The initial transmission attempt fails due to channel contention, but the retransmission eventually succeeds. The *RTS/CTS* mechanism is enabled, and the *minimum contention window* ($CW_{\min}$) is 7.

Draw a ***Timeline Diagram*** to represent the detailed sequence of events for the transmission of a MSDU from **Station *A* to Station *C*** based on the scenario described above.

Note:

- Use the *x-axis* for time and the *y-axis* to represent each of the three stations (*A, B,* and *C*) on separate horizontal lines.
- Show the contention behavior of all stations throughout the transmission cycle, including the randomly selected backoff counters and the range from which these counters are chosen.
- Represent key frame exchanges (*RTS, CTS, DATA, ACK*) with bold horizontal lines on the corresponding stations line. The line lengths should visually reflect relative transmission durations.
- Clearly indicate collisions, idle periods, *DIFS/SIFS* intervals, and successful transmission events, as dictated by the *IEEE 802.11 DCF* protocol.

Solution:



CamScanner

Figure 5: Time line diagram for MSDU Retransmission from station A to station C.

The diagram shown in Figure 5.

Key Events:

Note: This section is provided only for your clarification. It is not required as part of the answer.

1. **Idle Medium + DIFS** All stations sense idle medium → each waits for DIFS before starting backoff.
2. **Initial Backoff Selection (First Attempt, $n = 1$)** Backoff counter:
$$r \sim \text{Uniform}\{0, CW\}$$
where
$$CW_0 = CW_{\min} = 7 \quad \Rightarrow \quad r \in [0, 7]$$
Example picks: A=3, B=3, C=3 (slots). They count down during idle slots.
3. **First Transmission Attempt → Collision**
   - Station A reaches 0 first and sends RTS.
   - Station B also hits 0 → RTSRTS collision.
   - Station C also hits 0 → RTSRTS collision.
   - No CTS received → transmission failure.
4. **Exponential Backoff after Collision (Next Attempt, $n = 2$)** Binary exponential backoff (inclusive upper bound form):
$$CW_{k+1} = \min(2 \cdot CW_k + 1,\ CW_{\max})$$

Starting from $CW_0 = 7$:

$$CW_1 = \min(2 \cdot 7 + 1, CW_{\max}) = 15 \quad \Rightarrow \quad r \in [0, 15]$$

New random draws (example): A=4, B=6, C=9.

5. **Second Attempt → Success by A**
   - After DIFS, counters run. A hits 0 first → sends RTS(A→C).
   - After SIFS: CTS(C→A).
   - A sends DATA frame (longer duration).
   - After SIFS: ACK(C→A).
   - Medium goes idle → all stations re-enter contention.

3. a) A fleet of trucks is cruising down the highway in a tightly coordinated platoon. Suddenly, the lead truck detects an obstacle and initiates emergency braking. The safety of every following truck depends on how quickly they react. How does ***Vehicle-to-Vehicle (V2V) communication*** enhance emergency braking in platooning by improving safety, and increasing traffic flow efficiency compared to ***traditional systems without V2V***? 4.6 (CO1) (PO2)

Solution:



Figure 6: Emergency Braking in Truck Platooning.

Emergency Braking in Truck Platooning with V2V Communication

1. Manual Driving (Limitation)

- Drivers depend on human perception and reaction time before braking.
- This creates significant brake lag and delayed response.
- High risk of rear-end collisions when trucks follow closely.

2. Radar-based Collision Mitigation (Limitation)

- Onboard radar detects obstacles and braking actions.
- Faster than human response but still limited by sensor detection delay.
- Brake lag remains, restricting safe following distances.

3. V2V-enabled Platooning (Merits)

- Trucks exchange braking signals instantly via wireless V2V links.
- Synchronized braking removes perception and detection delays.
- Allows closer following distances with reduced collision risk.
- Enhances safety and improves traffic flow efficiency.
- Supports eco-driving by lowering drag and fuel consumption.

Note: The following section is provided only for your clarification. Either the above or the following section is required as part of the answer. Comparison Table

System	Limitation	Impact on Safety/Flow
Manual Driving	Human perception and reaction delay → large brake lag	Unsafe at close distances, higher collision risk
Radar-based Collision Mitigation	Sensor detection delay → brake lag persists	Safer than manual, but still limited following distances
V2V-enabled Platooning	Instant braking signal exchange, no detection delay	Safe close following, synchronized braking, improved efficiency

b) You are tasked with designing communication systems for two very different environments: 4 (CO2) (PO2)

- *Case 1:* A corporate office in a well-connected urban area that requires high-speed, reliable communication for real-time applications such as video conferencing and VoIP.
- *Case 2:* A remote, disconnected island where *intermittent* (i.e., available only at certain times) and low-bandwidth links are available. In this scenario, only non-real-time, best-effort communication (e.g., delayed messaging, data syncing) is required.

Evaluate the suitability of the *TCP/IP architecture* and the *Delay/Disruption Tolerant Network (DTN) architecture* for each of the above cases. Provide **detailed justification** for your choices.

Solution:

Case 1 Corporate office (urban, real-time apps like video calls/VoIP)

- **Best choice:** TCP/IP
- **Reason:**
  - Network is always connected, fast, and reliable → TCP/IP works well.
  - Supports real-time apps using QoS.

– DTN not suitable because store-and-forward adds extra delay.

Case 2 Remote island (intermittent, low-bandwidth, non-real-time apps)

- **Best choice:** DTN (Delay/Disruption Tolerant Network)
- **Reason:**
  - DTN can handle long delays and broken links with store-and-forward.
  - Ensures messages or data eventually get delivered even if connectivity is patchy.
  - TCP/IP not suitable since it assumes a stable end-to-end path, which does not exist here.

c) Consider a *Basic Service Set (BSS)* in a *Wireless LAN (WLAN)* that supports *Quality of Service (QoS)* features as defined in the ***IEEE 802.11e*** standard. The following devices are connected to this *WLAN*, each running a different application: 8 (CO2) (PO2)

- *Device A* is engaged in a **real-time voice call (VoIP)**,
- *Device B* is **streaming high-definition video content**,
- *Device C* is performing **general web browsing**.

All devices access the shared medium using ***Enhanced Distributed Channel Access (EDCA)***, a contention-based channel access mechanism under the *Hybrid Coordination Function (HCF)*. Explain how *EDCA* supports differentiated *QoS* among these devices.

Solution:

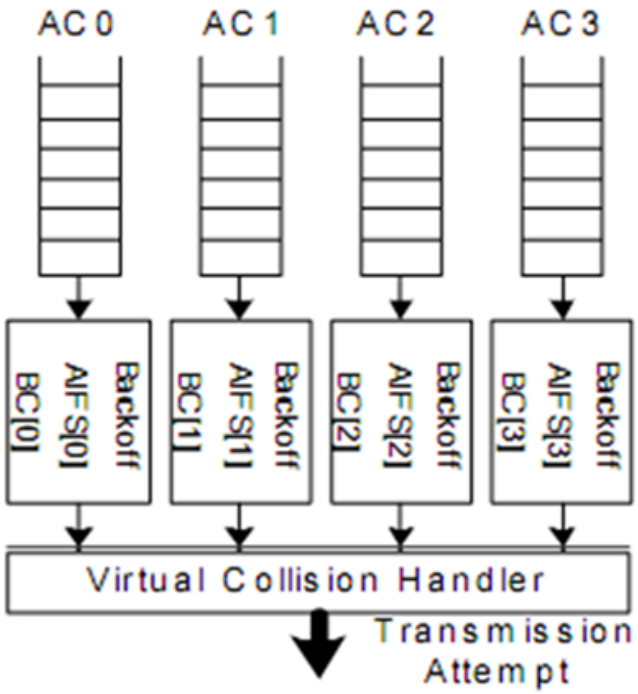


Figure 7: Backoff entities for every 802.11e stations

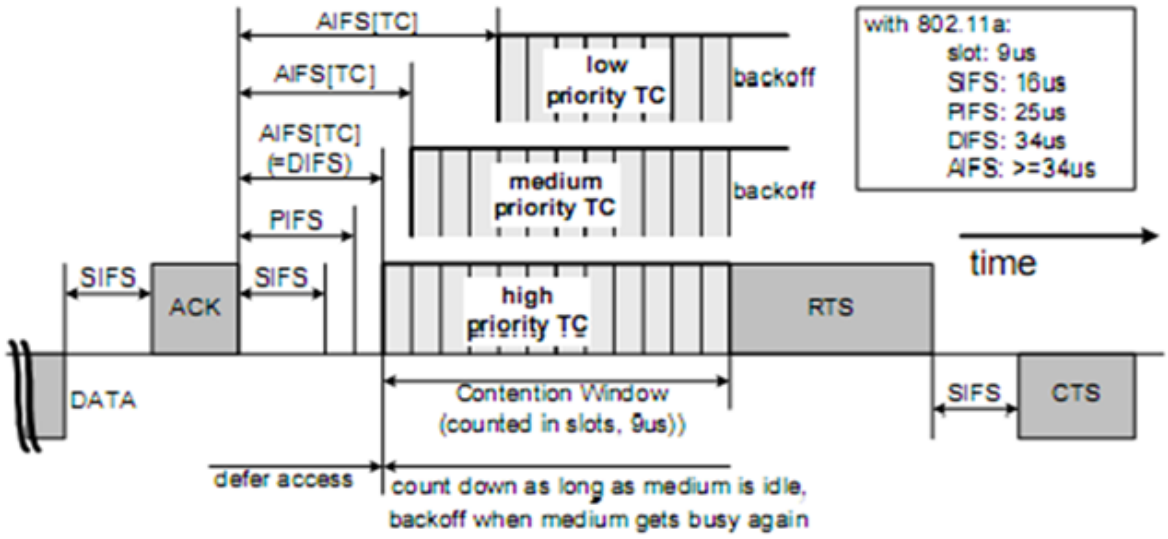


Figure 8: Sample contentionbased medium access in EDCA.

EDCA QoS Differentiation: AIFS and CW

Arbitration Interframe Space (AIFS)

Each Access Category (AC) uses a different Arbitration Interframe Space:

$$\text{AIFS[AC]} = \text{SIFS} + \text{AIFSN[AC]} \cdot \text{SlotTime}$$

- Smaller AIFS ⇒ Higher priority.
- **Voice (AC_VO):** shortest AIFS, highest priority.
- **Video (AC_VI):** slightly longer AIFS, second priority.
- **Best Effort (AC_BE)** and **Background (AC_BK):** longest AIFS, lowest priority.

Contention Window (CW)

The size of the contention window in backoff stage i after $i-1$ failed attempts is:

$$CW_i[AC] = \min\left[2^i \cdot (CW_{\min}[AC] + 1) - 1,\ CW_{\max}[AC]\right]$$

- Smaller $CW_{\min}$ and $CW_{\max}$ ⇒ Higher access probability.
- **AC_VO (voice):** smallest CW, very fast access.
- **AC_VI (video):** small CW, but slightly larger than AC_VO.
- **AC_BE and AC_BK:** larger CW, longer waiting time, lower priority.

QoS Effect Across Devices

- **Device A (VoIP, AC_VO):** very short AIFS + smallest CW ⇒ low delay, low jitter.
- **Device B (Video streaming, AC_VI):** medium AIFS, small CW ⇒ good throughput.
- **Device C (Web browsing, AC_BE):** longer AIFS + larger CW ⇒ waits more, only uses leftover capacity.

Note: This section is provided only for your clarification. Summary Table:)

Access Category (AC)	**AIFS**	**CW Range**
AC_VO (Voice)	Shortest (highest priority)	Smallest CW_{min}, CW_{max}
AC_VI (Video)	Short (medium priority)	Small CW_{min}, CW_{max}
AC_BE (Best Effort)	Longer	Larger CW_{min}, CW_{max}
AC_BK (Background)	Longest (lowest priority)	Largest CW_{min}, CW_{max}